Orion68-CPU-2

File: orion68cpu2.kicad_sch

Orion68-MEM

File: orion68mem.kicad_sch

Orion68-IO

File: orion68io.kicad_sch

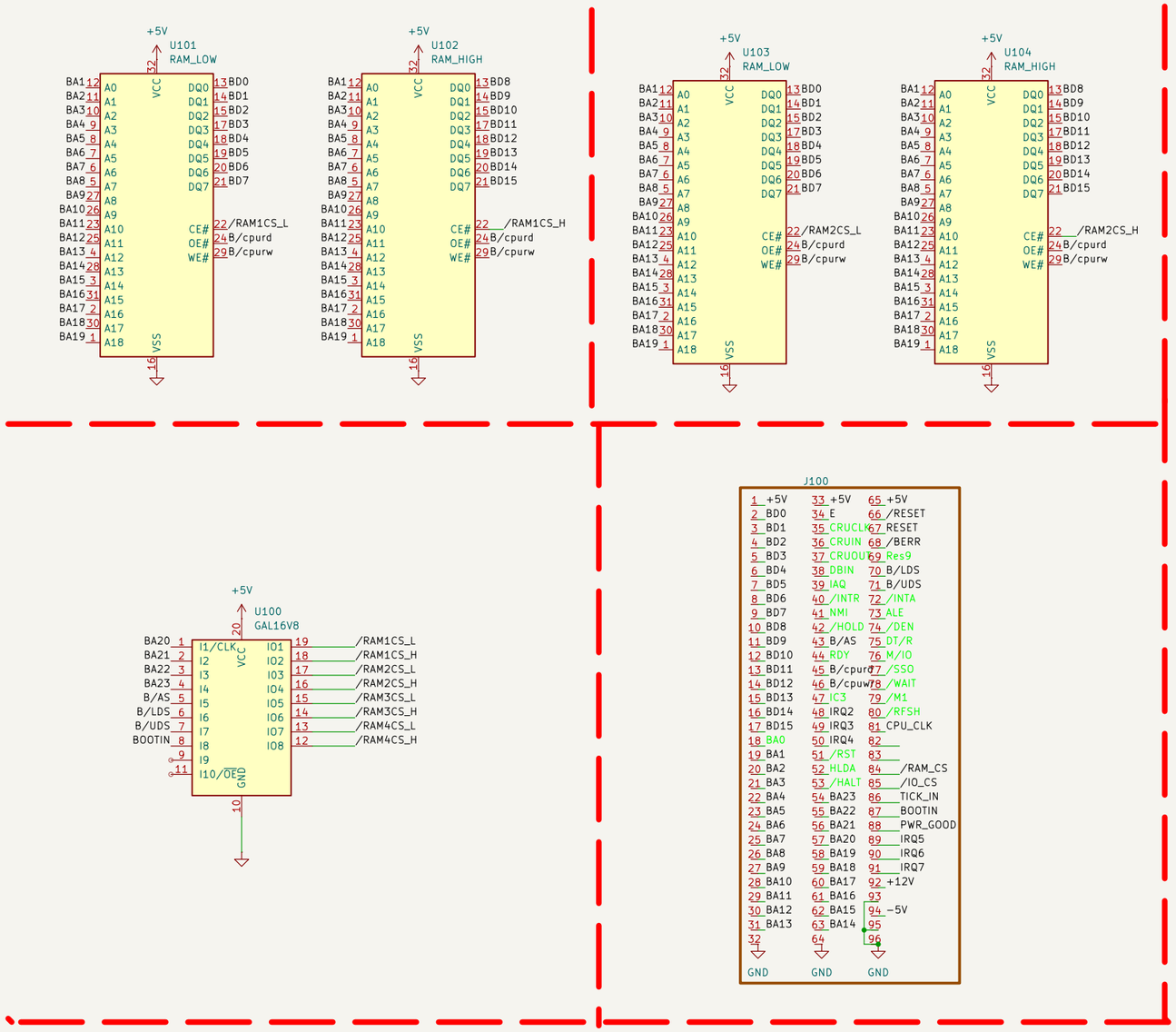
Orion68-VID

File: orion68vid.kicad_sch

Orion68 Project

Orion68 Project

2026-03-14



Memory board – 100

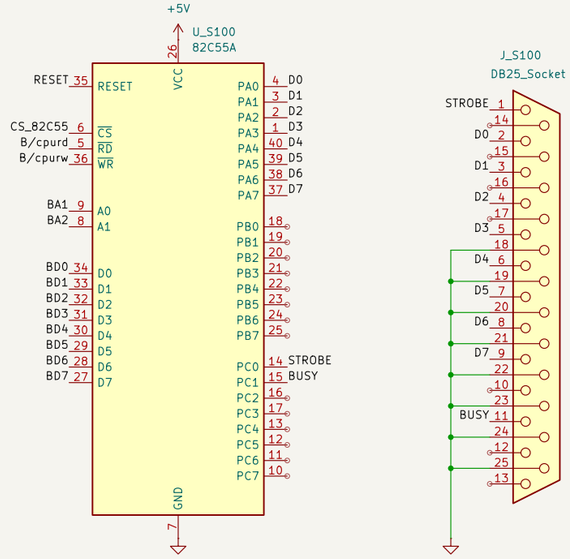
Orion68-MEM

2026-03-14

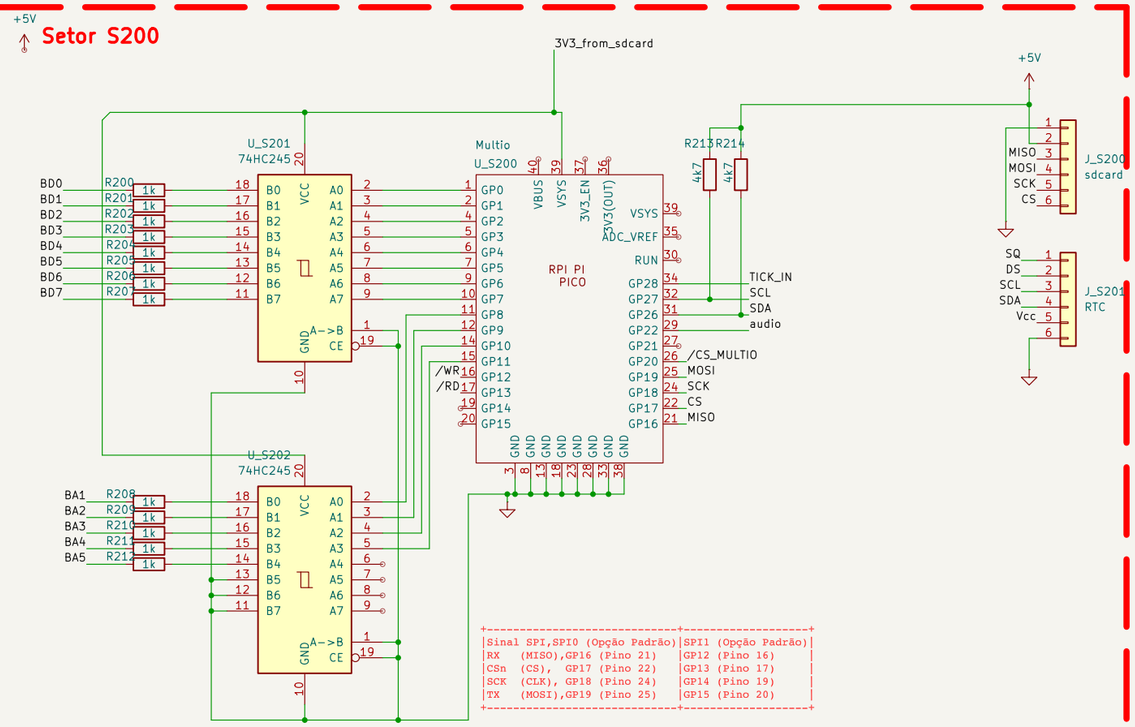
Setor S100

DB25 fêmea

Pino	Nome	Direção (PC)	Função
1	Strobe	Saída	Indica que os dados são válidos (Ativo em 0)
2	D0	E/S	Bit de Dados 0 (LSB)
3	D1	E/S	Bit de Dados 1
4	D2	E/S	Bit de Dados 2
5	D3	E/S	Bit de Dados 3
6	D4	E/S	Bit de Dados 4
7	D5	E/S	Bit de Dados 5
8	D6	E/S	Bit de Dados 6
9	D7	E/S	Bit de Dados 7 (MSB)
10	Ack	Entrada	Reconhecimento de recebimento (Ativo em 0)
11	Busy	Entrada	Impressora ocupada
12	Paper Out	Entrada	Fim de papel
13	Select	Entrada	Impressora online
14	Auto Linefeed	Saída	Avanço de linha automático
15	Error	Entrada	Erro na impressora
16	Init	Saída	Inicializar impressora
17	Select In	Saída	Selecionar impressora
18-25	GND	---	Terra (Ground)

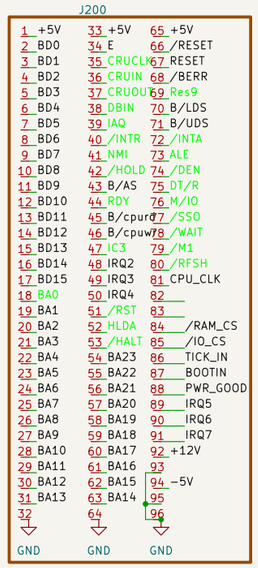


Setor S200

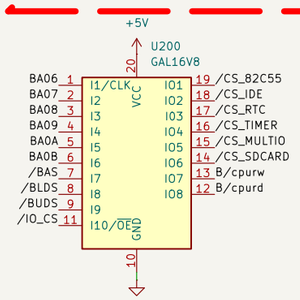


Sinal SPI,SPI0 (Opção Padrão)	SPI1 (Opção Padrão)
RX (MISO),GP16 (Pino 21)	GP12 (Pino 16)
CSn (CS), GP17 (Pino 22)	GP13 (Pino 17)
SCK (CLK), GP18 (Pino 24)	GP14 (Pino 19)
TX (MOSI),GP19 (Pino 25)	GP15 (Pino 20)

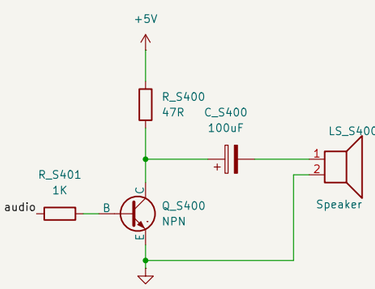
Binário A5...A1	LDS A0	Ender. Decimal	D0-D7 (LDS) D8-D15 (UDS)
00000	0	0(Par)	LDS
00000	1	1(Ímpar)	UDS
00001	0	2(Par)	LDS
00001	1	3(Ímpar)	UDS
00010	0	4(Par)	LDS
00010	1	5(Ímpar)	UDS
00011	0	6(Par)	LDS
00011	1	7(Ímpar)	UDS
00100	0	8(Par)	LDS
00100	1	9(Ímpar)	UDS
00101	0	10(Par)	LDS
00101	1	11(Ímpar)	UDS
00110	0	12(Par)	LDS
00110	1	13(Ímpar)	UDS
00111	0	14(Par)	LDS
00111	1	15(Ímpar)	UDS
01000	0	16(Par)	LDS
01000	1	17(Ímpar)	UDS
01001	0	18(Par)	LDS
01001	1	19(Ímpar)	UDS
01010	0	20(Par)	LDS
01010	1	21(Ímpar)	UDS
01011	0	22(Par)	LDS
01011	1	23(Ímpar)	UDS
01100	0	24(Par)	LDS
01100	1	25(Ímpar)	UDS
01101	0	26(Par)	LDS
01101	1	27(Ímpar)	UDS
01110	0	28(Par)	LDS
01110	1	29(Ímpar)	UDS
01111	0	30(Par)	LDS
01111	1	31(Ímpar)	UDS
10000	0	32(Par)	LDS
10000	1	33(Ímpar)	UDS
10001	0	34(Par)	LDS
10001	1	35(Ímpar)	UDS
10010	0	36(Par)	LDS
10010	1	37(Ímpar)	UDS
10011	0	38(Par)	LDS
10011	1	39(Ímpar)	UDS
10100	0	40(Par)	LDS
10100	1	41(Ímpar)	UDS
10101	0	42(Par)	LDS
10101	1	43(Ímpar)	UDS
10110	0	44(Par)	LDS
10110	1	45(Ímpar)	UDS
10111	0	46(Par)	LDS
10111	1	47(Ímpar)	UDS
11000	0	48(Par)	LDS
11000	1	49(Ímpar)	UDS
11001	0	50(Par)	LDS
11001	1	51(Ímpar)	UDS
11010	0	52(Par)	LDS
11010	1	53(Ímpar)	UDS
11011	0	54(Par)	LDS
11011	1	55(Ímpar)	UDS
11100	0	56(Par)	LDS
11100	1	57(Ímpar)	UDS
11101	0	58(Par)	LDS
11101	1	59(Ímpar)	UDS
11110	0	60(Par)	LDS
11110	1	61(Ímpar)	UDS
11111	0	62(Par)	LDS
11111	1	63(Ímpar)	UDS



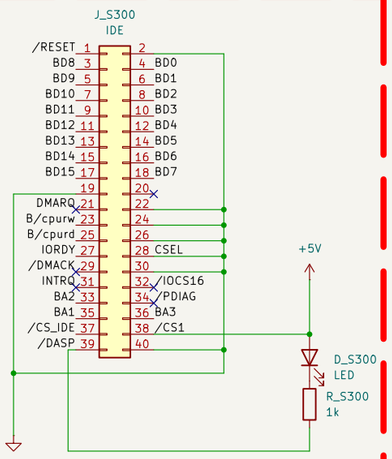
Setor S400




```
0xF00040-0xF0007F -> 82C55
0xF00080-0xF000BF -> IDE
0xF000C0-0xF000FF -> RTC
0xF00140-0xF0017F -> TIMER
0xF00180-0xF001BF -> SDCARD
0xF001C0-0xF001FF -> free
0xF00240-0xF0027F -> free
0xF00280-0xF002BF -> free
0xF002C0-0xF002FF -> free
0xF00340-0xF0037F -> free
0xFB8000-0xFB803F -> VGA-CARD
```



Setor S300

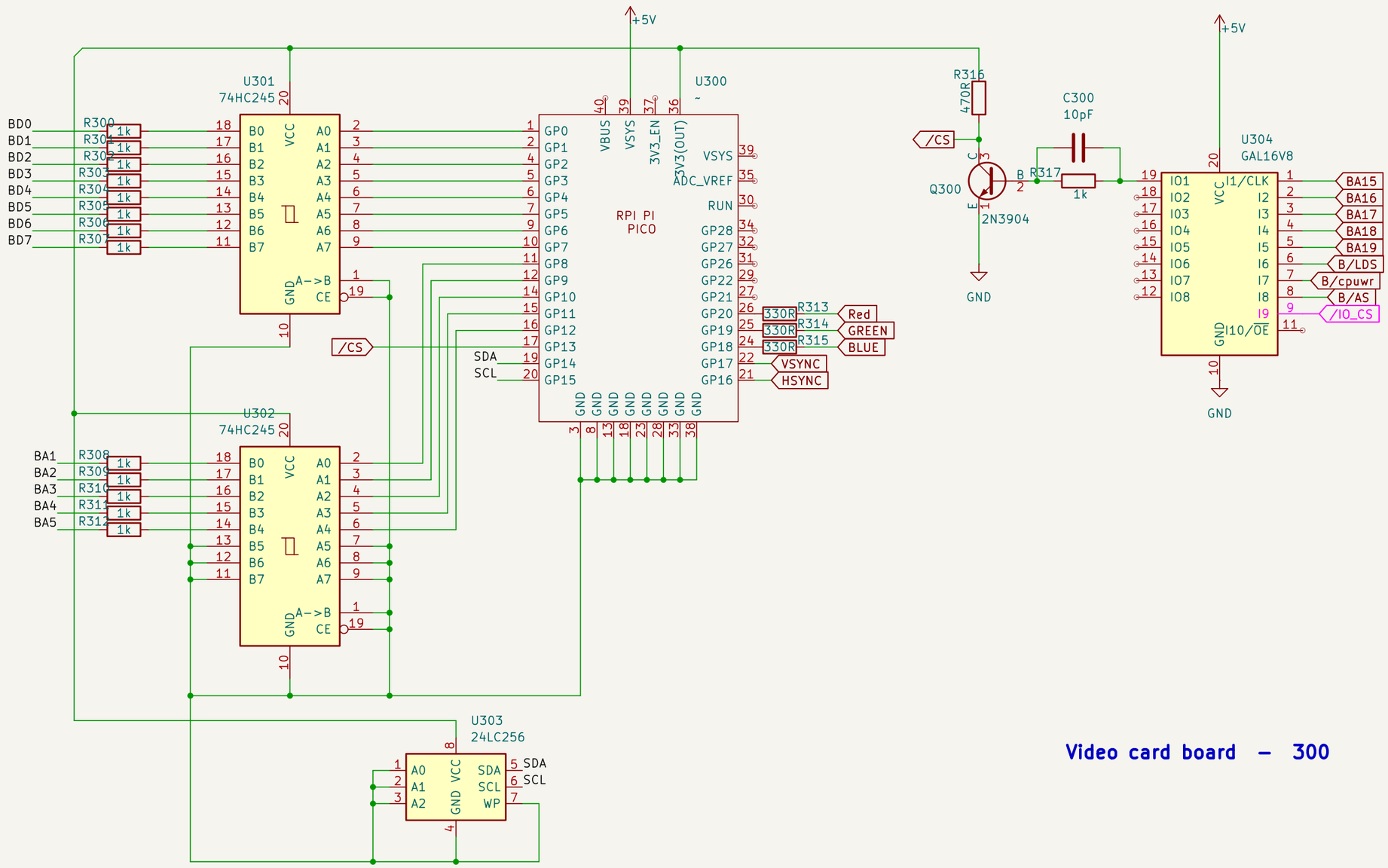


IO board – 200

Orion68-IO

2026-03-14

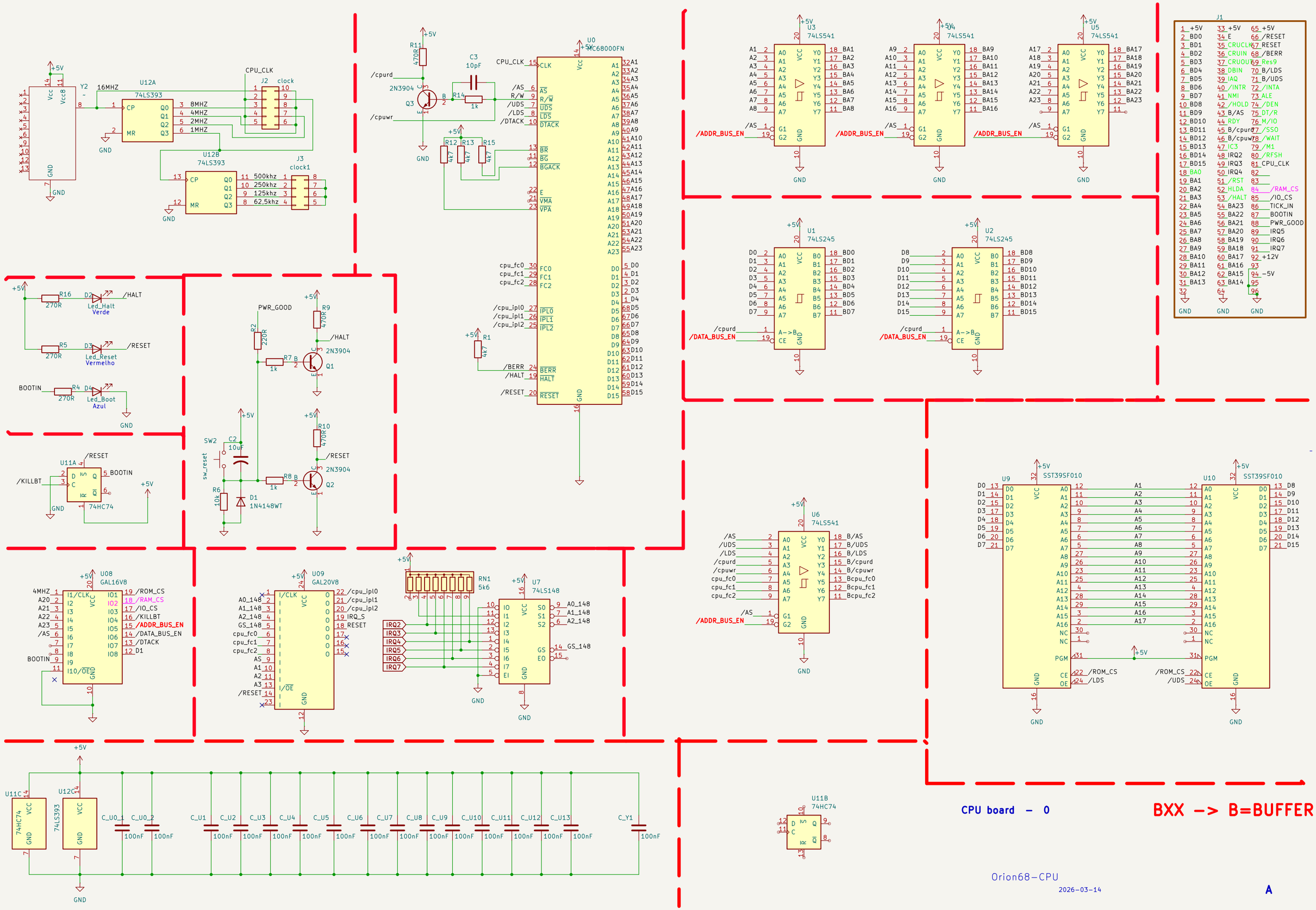
```
19 18 17 16 | 15
 1  0  0  0 |  0 = F80000
 1  0  0  0 |  1 = F88000
 1  0  0  1 |  0 = F90000
 1  0  0  1 |  1 = F98000
 1  0  1  0 |  0 = FA0000
 1  0  1  0 |  1 = FA8000
 1  0  1  1 |  0 = FB0000
 1  0  1  1 |  1 = FB8000
pino 1 (A0)  = A15 do 68000
pino 2 (A1)  = A16 do 68000
pino 3 (A2)  = A17 do 68000
pino 4 (/E0) = A18 do 68000
pino 5 (/E1) = /LDS R/W /AS do 68000
pino 6 (E2)  = A19 do 68000

com isso a saida pino 7 /Y7 só deve ser acessada com 0xB8000
```



Video card board – 300

Orion68-VID

2026-03-14


U0
MC68000FN
U12A
74LS393
U12B
74LS393
J2 clock
J3 clock1
CPU_CLK
16MHZ
Y2
U3
74LS541
U4
74LS541
U5
74LS541
U1
74LS245
U2
74LS245
U6
74LS541
U08
GAL16V8
U09
GAL20V8
U7
74LS148
RN1
5k6
U9
SST39SF010
U10
SST39SF010
U11A
74HC74
U11B
74HC74
U11C
74HC74
Q3
2N3904
Q1
2N3904
Q2
2N3904
D1
1N4148WT
SW2
sw_reset
Led_Halt
Verde
Led_Reset
Vermelho
Led_Boot
Azul
/ADDR_BUS_EN
/DATA_BUS_EN
J1
CPU board – 0
BXX -> B=BUFFER
Orion68-CPU
2026-03-14
A